

ACCESS

ACCESS CONTROL AND CREDENTIAL EXCHANGE FOR SPACE SYSTEMS

ESTABLISH TRUST. EXCHANGE CREDENTIALS.
ENFORCE BOUNDED ACCESS. AUTONOMOUSLY.

NASA / CIGRUS AT THE INTERNATIONAL SPACE STATION

THE QUESTION / 01

MAY THIS INDEPENDENTLY OPERATED
VEHICLE PERFORM THIS SPECIFIC
ACTION, HERE, NOW, UNDER THESE
CONSTRAINTS?

THE GAP / 02

SPACE IS STANDARDIZING HOW
SYSTEMS CONNECT, BUT NOT YET
HOW THEY TRUST AND
AUTHORIZE ONE ANOTHER.

EXISTING LAYER	EXISTING LAYER	ACCESS PROTOCOL
SECURE THE LINK CCSDS security protects commands and data between established endpoints.	ENABLE THE MANEUVER RPO, navigation, docking standards, certification, and safety logic enable compatible maneuvers.	AUTHENTICATE AND AUTHORIZE ACCESS verifies peer identity and grants bounded, single-use authority for a specific action under current conditions.

Crew Dragon, Mission Extension Vehicles, and ELSA-d show that autonomous rendezvous, docking, capture, and servicing are operational today. Yet trust and permission still rely on tightly integrated missions and bilateral agreements. As operators, spacecraft, and robots multiply, authentication and authorization must become autonomous.

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ISAM CAPABILITY ENTRIES CATALOGED BY NASA IN 2025, ACROSS 145 DEVELOPER PROFILES

THE PROTOCOL FLOW / 03

AUTHENTICATE THE ENCOUNTER.
THEN AUTHORIZE THE ACTIONS.

VISITING SPACECRAFT: ACCESS CLIENT (AC) / HOST SPACECRAFT: AUTHORITY (AAS), POLICY (APS), EVIDENCE (AEP), ENFORCEMENT (AEG)

01	02	03	04	05
REQUEST SESSION The visitor asks the host to begin an ACCESS session.	TRUST THE ENCOUNTER The host verifies credentials and applies policy.	REQUEST ACTION The visitor requests one protected action.	DECIDE AND ISSUE The host issues a signed, single-use entitlement.	USE ENTITLEMENT The visitor presents it; the host gateway consumes it and permits the action.

THE DECISION IS MADE ONBOARD. NO
REAL-TIME GROUND CONTACT REQUIRED.

OPERATOR-DEFINED POLICY / CURRENT EVIDENCE / ONBOARD SAFETY CONSTRAINTS

WHAT ACCESS STANDARDIZES

THE CONTRACT FOR
INTEROPERABILITY.

- + Roles, trust anchors, and authority boundaries

+ Schemas, canonical encoding, and credential bindings
- + Protocol states, sequencing, timeouts, errors, and recovery

+ Security profiles, version negotiation, and conformance requirements

STANDARDS DEVELOPMENT

A PATH TO AN
INTEROPERABLE
STANDARD.

ACCESS development combines a working protocol model with specification and interoperability work. Technical progress alone is not enough. Spacecraft operators, vehicle and infrastructure providers, integrators, mission-assurance experts, and standards organizations must help shape requirements, validate the approach, and support adoption.

- DOCKING
- REFUELING
- CAPTURE
- SERVICING
- ASSEMBLY

DEPOT ACCESS

THE OBJECTIVE

A COMMON TRUST PROTOCOL.
INDEPENDENT CONTROL.

Interoperable authentication and bounded authorization, while each operator retains control of policy, safety, and enforcement.